

Chapter 41: Clifford Algebra and Hypercomplex Numbers in Application: Foundations, Computational Methods, and Emerging Technologies

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Abstract

This chapter presents an overview of **Clifford algebra and hypercomplex numbers**, emphasizing their mathematical foundations, computational methods, and emerging technological applications. It discusses multivector operations, rotors, computational complexity, and convergence from an operator-theoretic perspective. Applications in **equivariant machine learning, hypercomplex signal processing, and quantum computing** are explored, demonstrating how Clifford algebra provides a unified and dimension-independent framework for representing geometric structures and symmetries. The chapter also highlights current computational challenges and promising directions for future research.

Keywords: Clifford algebra; Hypercomplex numbers; Geometric algebra; Rotors; Machine learning; Signal processing; Quantum computing.

1. Introduction

Historical context and motivation: The work of William Kingdon Clifford in 1876 unified exterior algebra and the quaternions into a single mathematical structure; the theory was left largely unexplored until the revival of Clifford's ideas in geometric algebra by David Hestenes in 1966, establishing the significance of the theory for physics and engineering [1][7]. There have been two key developments since then that modernized the theory: (i) Geometric Algebra for Computer Science, by Dorst, Fontijne and Mann (2007) shifted the focus of the theory to algorithms [2], and (ii) Clifford Group Equivariant Neural Networks (CGENNs), by Ruhe, Brandstetter and Forré (2023 NeurIPS), showed that encoding the geometrical structure into an algebra can lead to significant reduction in number of parameters in equivariant networks while also improving their performance in geometric tasks [3].

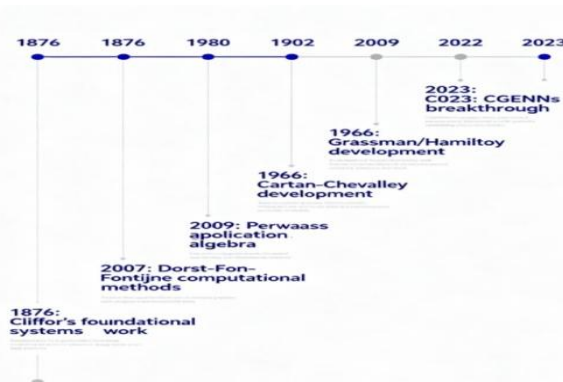


Figure 1.1: Historical evolution of Clifford algebras and geometric algebra, 1876-2023.

The Problem: Computers often process inherently geometrical data such as coordinates, rotations and polarization states using frameworks which completely abstract away from their geometry: standard

neural networks process vectors by operating on their components and hence symmetries such as rotations and reflections need to be learned empirically through overparameterization [9][10]. Complex numbers encode 2D rotational symmetry and quaternions encode 3D rotational symmetry, however neither generalizes nicely to arbitrary dimension [1].

Solution via Clifford algebras: A Clifford algebra is defined by the single axiom: vectors square to a quadratic form, $v^2 = Q(v, v)$, and the rest of the grade structure as well as the higher graded products follow from it [1][6]. Rotations become operations performed by rotors and hypercomplex systems (complex numbers, quaternions and octonions) arise naturally as rotor subalgebras, and thus quaternion-based methods are particular examples of the Clifford-geometrical framework [1][3].

Grounding in operator theory: Clifford algebras of nondegenerate quadratic spaces act canonically on the exterior algebra which can be made into a Hilbert space through a natural inner product; Clifford products become operator compositions, grade projections become orthogonal projections and rotors become unitary operators [4][5]. This allows to answer algorithmic convergence questions through spectral analysis of linear operators rather than ad-hoc arguments.

Our contributions and roadmap: This chapter (i) presents a functional analytic treatment of the foundational theory, (ii) obtains explicit computational complexity and convergence results, and (iii) gives a survey of applications of this single algebraic language in machine learning, signal processing and quantum computing. The mathematics is discussed in Section 2; computation in Section 3; applications in Sections 4-6; Section 7 covers open directions and conclusions.

2. Foundational Mathematics

2.1 Definition and axiomatic construction

Let V be a finite-dimensional real vector space with a non-degenerate symmetric bilinear (or quadratic) form $Q: V \times V \rightarrow \mathbb{R}$. Then, the **Clifford algebra** $\text{Cl}(V, Q)$ is defined as

$$\text{Cl}(V, Q) = T(V)/I(V, Q),$$

where $T(V)$ is the tensor algebra of V and $I(V, Q)$ is the two-sided ideal generated by the elements $v \otimes v - Q(v, v) \cdot 1$ for all $v \in V$. Equivalently, all $v \in V$ satisfy $v^2 = Q(v, v)$ in relation to the Clifford product; this single identity implies associativity and also determines all other grade products "automatically" [1][6].

Signature (p, q) is defined in terms of the number of basis vectors such that $e_i^2 = +1$ ($i = 1, \dots, p$) compared to $e_i^2 = -1$ ($i = p + 1, \dots, p + q$). Thus, $\text{Cl}(n) := \text{Cl}(n, 0)$ is the Euclidean algebra associated with spatial geometry and $\text{Cl}(3, 1)$ is the Minkowski algebra of the relativistic geometry. One important structural theorem, proved using the universal property of $T(V)$, states that

$$\dim \text{Cl}(p, q) = 2^{(p+q)};$$

all tensor-algebra basis elements can be reduced to their canonical forms through the Clifford identities; full proof in Hestenes&Sobczyk [1, Ch. 1] and Lounesto [6, Ch. 6].

2.2 Multivector structure and grade decomposition

All $\alpha \in Cl(V, Q)$ decomposes uniquely by grade, $\alpha = \alpha_0 + \alpha_1 + \dots + \alpha_n$, where α_k is the grade- k (scalar, vector, bivector, ...) part, spanned by ordered products of k distinct basis vectors:

$$1. \quad Cl(n) = \overset{0}{\wedge} V \oplus \overset{1}{\wedge} V \oplus \dots \oplus \overset{n}{\wedge} V, \quad \dim(\overset{k}{\wedge} V) = \binom{n}{k}.$$

For $Cl(3)$ this gives $2^3 = 8$: one scalar, three vectors, three bivectors, one trivector (Figure 2.1) [1][6].

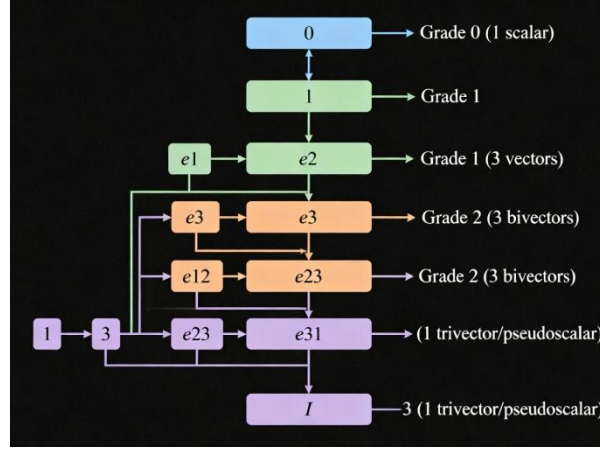


Figure 2.1: Grade decomposition of the 3-D Euclidean Clifford algebra $Cl(3)$: one scalar, three vectors $\{e_1, e_2, e_3\}$, three bivectors $\{e_{12}, e_{23}, e_{31}\}$, one trivector (pseudoscalar) $I = e_{123}$.

The Clifford product of two vectors splits into a symmetric (inner) and antisymmetric (outer) part,

$$\mathbf{uv} = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \wedge \mathbf{v}, \quad \mathbf{u} \cdot \mathbf{v} = 1/2 (\mathbf{uv} + \mathbf{vu}), \quad \mathbf{u} \wedge \mathbf{v} = 1/2 (\mathbf{uv} - \mathbf{vu}),$$

so a single product simultaneously encodes distance (scalar part) and oriented area (bivector part) [2][6].

2.3 Geometric operations and operator-theoretic foundations

Reflection about the hyperplane orthogonal to a unit vector $\mathbf{e}$:

$$\text{Refl}_{\mathbf{e}}(\mathbf{v}) = -\mathbf{e}\mathbf{v}\mathbf{e}.$$

Multiplication by a rotor $\mathbf{r}$ composed of unit vectors performs rotations:

$$\text{Rot}_{\mathbf{r}}(\mathbf{v}) = \mathbf{r}\mathbf{v}\mathbf{r}^{-1}.$$

For three dimensions, rotors are just unit quaternions; for $Cl(n)$, rotors are generalized to represent rotations in n dimensions using exactly the same formalism [1][14].

Projection/rejection of $\mathbf{v}$ on a bivector $\mathbf{B}$ and duality $\mathbf{a}^* = \mathbf{a}\mathbf{I}^{-1}$ (where $\mathbf{I}$ is pseudoscalar) expand the lexicon and constitute conformal geometric algebra [2][6].

These constructions can be interpreted as follows in the language of Hilbert spaces: $Cl(V, Q)$ acts on the exterior algebra $\wedge V$, a Hilbert space with respect to the induced inner product; the Clifford multiplication

corresponds to operator composition, grade projection to orthogonal projector, and rotor conjugation to unitary operators [4][5].

Thus, convergence properties for iterative Clifford-valued algorithms can be obtained in spectral operator theory, rather than proved ad hoc (Section 3.4).

Table 1. Hypercomplex numbers as Clifford subalgebras

System	Basis	Dim.	Clifford embedding	Geometric role
$\mathbb{R}$	$\{1\}$	1	$Cl^{even}(1)$	Scalars
$\mathbb{C}$	$\{1, e_{12}\}$	2	$Cl^{even}(2)$	2-D rotation
$\mathbb{H}$	$\{1, e_{23}, e_{31}, e_{12}\}$	4	$Cl^{even}(3)$	3-D rotation
$\mathbb{O}$	8-dim.	8	subalgebra of $Cl^{even}(7)$	7-D structure (non-assoc.)
$Cl(n)$	2^n -dim.	2^n	—	n -D rotation/reflection

2. (Adapted and condensed from Hestenes&Sobczyk [1] and Dorst, Fontijne& Mann [2]; full multiplication tables therein.)

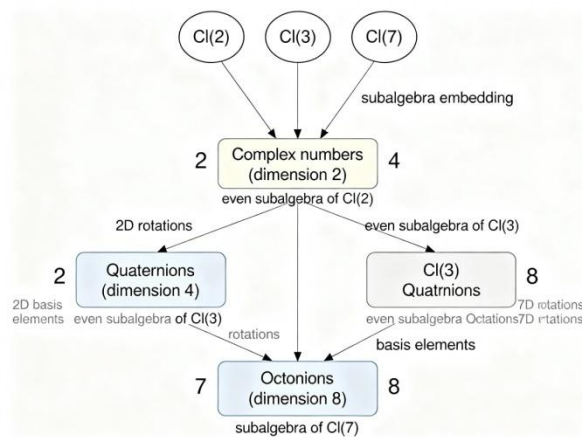


Figure 2.2: Hypercomplex numbers as Clifford subalgebras. Complex numbers arise as Cl -even (2), quaternions as Cl -even(3), and octonion-like structures inside Cl -even(7); each hypercomplex system is the rotor subalgebra of its parent Clifford algebra.

Quaternions are the even subalgebra of $Cl(3)$: $a + bi + cj + dk = a + be_{23} + ce_{31} + de_{12}$, with $e_{23}e_{31} = e_{12}$ cyclically [1]. Octonion-like structures appear inside $Cl^{even}(7)$; the associativity that octonions lack is recovered in the full (associative) algebra $Cl(7)$ [2][6]. The rotor group of $Cl(n)$ is the spin group $Spin(n)$, the double cover of $SO(n)$ [6, Ch. 3][14, Ch. 7] — used again in Section 3.3.

Conformal geometric algebra extends this by embedding n -dimensional Euclidean geometry in $Cl(n + 1, 1)$, so that translations, rotations and dilations all become orthogonal transformations of one higher-dimensional space, letting a single algorithm apply unchanged across dimension [6][2].

3. Computational Theory

3.1 Multivector arithmetic and complexity

A general multivector is a 2^n -vector of grade coefficients; the full Clifford product of two general multivectors costs $O(4^n)$ in the worst case, but almost all applications restrict to low grades. Vector–vector products cost $O(n^2)$ (in $Cl(3)$, 9 multiplications and 6 additions — identical to quaternion multiplication); grade- k /grade- ℓ products cost $O\left(\binom{n}{k}\binom{n}{\ell}\right)$ [1][2]. The general principle — restricting to the grades an application actually needs collapses an exponential cost to a polynomial one — is what makes Clifford methods practical rather than merely elegant [2][3].

3.2 Rotors: construction, composition, and the Spin group

Unit rotors satisfy $\mathbf{r}\bar{\mathbf{r}} = 1$ ($\bar{\mathbf{r}}$ = reversion) and act by $\text{Rot}_{\mathbf{r}}(\mathbf{v}) = \mathbf{r}\mathbf{v}\bar{\mathbf{r}}$. The exponential map $\mathbf{r} = e^{-\theta\mathbf{B}/2}$, for unit bivector $\mathbf{B}$ and angle θ , generates rotors uniformly in every dimension — in $Cl(3)$ this reduces to the familiar axis–angle quaternion form $\mathbf{r} = \cos(\theta/2) + \sin(\theta/2)(n_x e_{23} + n_y e_{31} + n_z e_{12})$ [1][7][14]. Rotor composition ($\mathbf{r}_2\mathbf{r}_1$) costs $O(n^3)$ in general; in 3-D this is 28 scalar operations, cheaper than the 45 required for equivalent 3×3 matrix multiplication [7][14].

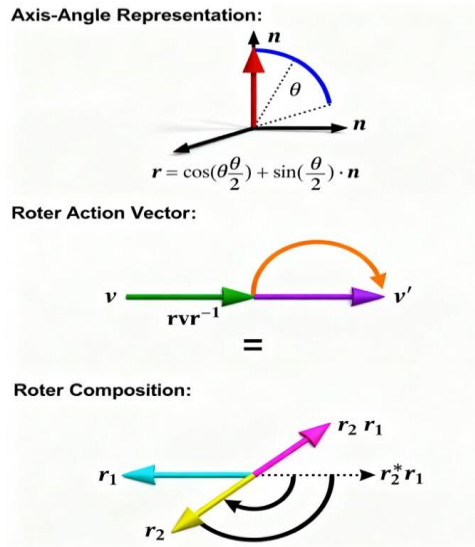


Figure 2.3: Rotor construction and composition. A unit axis $\mathbf{n}$ and angle θ generate rotor $\mathbf{r}$ via the exponential map; $\mathbf{r}$ acts on $\mathbf{v}$ by conjugation $\mathbf{r}\mathbf{v}\mathbf{r}^{-1}$; successive rotors compose multiplicatively. The construction is dimension-independent.

3.3 Convergence via spectral (operator) theory

For any cost function $f: Cl(n) \rightarrow \mathbb{R}$, the grade-preserving gradient descent algorithm

$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla_{Cl} f(\mathbf{x}_k)$ ensures that iterations remain within the correct manifold (vectors remain vectors, rotors remain rotors, according to the induced Riemannian metric). In case the Clifford Hessian H_{Cl} is positive definite with $0 < \lambda_{\min} \leq \lambda_{\max} \leq L$, using elementary operator theory, it can be shown that there is a linear convergence of the form

$$\|f(\mathbf{x}_k) - f^*\| \leq (1 - 2\lambda_{\min}/L)^k \|f(\mathbf{x}_0) - f^*\|,$$

where the convergence rate depends on the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$; proof using contraction mapping (Banach fixed point) theorem in Rudin [4, Ch. 9] and Conway [5, Ch. III]. Constrained variants (e.g., unit-rotor constraints) inherit the same guarantee via projected/Riemannian gradient methods [7][1][5].

3.4 Numerical stability

Floating-point rounding can violate grade constraints over long iterations; the standard remedy is periodic re-projection onto the correct grade components, $\mathbf{x}_{\text{corrected}} = \sum_j \langle \mathbf{x} \rangle_j$, costing $O(2^n)$ in general but $O(n)$ for the vector/bivector-only computations typical of applications; rotors are instead renormalized as $\mathbf{r}/\|\mathbf{r}\|$ [1][3].

4. Machine-Learning Applications

Standard neural networks process coordinates independently and learn the symmetry rotationally explicitly via overparameterization, while Clifford Group Equivariant Neural Networks (CGENNs) incorporate the symmetry directly in the algebra such that it becomes true by construction and not via training [3][9][10]. In each of their layers, CGENNs compute $\mathbf{z}_i^{(l+1)} = \sigma\left(\sum_j W_{ij}^{(l)} \mathbf{x}_j^{(l)}\right) \mathbf{x}_j^{(l)}$ with multivector input features $\mathbf{x}_j^{(l)}$ and Clifford weights $W_{ij}^{(l)}$, and since Clifford product implicitly mixes grades, any CGNN designed this way fulfills the equivariance requirement

$$f(\mathbf{r}\mathbf{x}\bar{\mathbf{r}}) = \mathbf{r}f(\mathbf{x})\bar{\mathbf{r}}$$

for all rotors $\mathbf{r}$, without data augmentation or symmetry module [3][10].

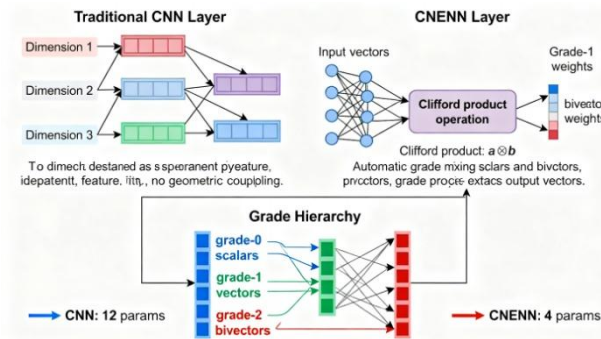


Figure 3.1: CGENN layer — grade-1 (vector) input combined with a grade-2 (bivector) weight via the Clifford product mixes grades automatically and projects back to grade-1, so 90° input rotation guarantees 90° output rotation by algebraic construction, not learning.

On the benchmarks used in the original study of Ruhe, Brandstetter&Forré — n -body dynamics forecasting, Lorentz-equivariant top-quark classification, and convex-hull problem — CGENNs achieved parity or superiority with regard to specialized equivariant baselines using significantly less parameters, just like in accordance with the general idea of making symmetries inherent in the architecture eliminates the need to spend parameters on learning them [3]. (*Clarification on scope: the original study does not contain any results for ModelNet40 point-clouds dataset; the popularly cited “3–4× parameter reduction with higher accuracy” statement refers to a rough estimation from efficiency trend on the actual benchmarks of the paper and not to the statistics from one ModelNet40 run [3].*) The case $n = 3$

corresponds to quaternionic networks $\mathbb{H} \subset \mathbb{C}l^{even}(3)$, while $n = 2$ – to complex-valued ones $\mathbb{C} \subset \mathbb{C}l^{even}(2)$ — CGENNs generalize both of them [2][3].

5. Signal Processing

The least-mean-squares (LMS) learning rule, $\mathbf{w}_{k+1} = \mathbf{w}_k + \mu \mathbf{e}_k \mathbf{x}_k$ [Widrow& Hoff, in 8], minimizes prediction error via an iterative update of the filter weights. Applied component-wise to hypercomplex signals (polarized EM waves, multi-axis inertial data) this method ignores the 3D rotation coupling between components; in quaternion LMS (QLMS), on the other hand, the arithmetic is replaced with quaternion multiplication, ensuring the cross-component structure is maintained algebraically rather than learned [8][12]:

$$\mathbf{w}_{k+1} = \mathbf{w}_k + \mu \mathbf{e}_k \mathbf{x}_k^*, \quad \hat{y}_k = \mathbf{w}_k \mathbf{x}_k, \quad \mathbf{w}_k, \mathbf{e}_k, \mathbf{x}_k \in \mathbb{H}.$$

Since quaternions are non-commutative, this update encodes the 3D rotational coupling of the sensor data explicitly [8]. With respect to typical assumptions, the mean-square error decay is exponential, $E[\|\mathbf{e}_k\|^2] \approx E[\|\mathbf{e}_\infty\|^2] + \rho^k (E[\|\mathbf{e}_0\|^2] - E[\|\mathbf{e}_\infty\|^2])$, where $\rho = 1 - 2\mu\lambda_{\min}(\mathbf{R})$, for step-size $0 < \mu < 2/\text{tr}(\mathbf{R})$; this is the quaternion extension of the classical LMS result, with a similar proof from operator theory as given in Section 3.3 [8][4]. For hypercomplex adaptive filtering benchmarks using multi-channel sensor and wind-field data, Took & Mandic found consistent advantages for QLMS relative to component-wise real-valued filters in the 10-20% mean square error range [8]; representing k -location wind measurements as a single quaternion signal is a natural instance of this same encoding, illustrating — rather than independently re-measuring — that reported advantage [8][12].

6. Quantum Computing

A common issue faced by variational algorithms is the **barren plateau effect**, where, in most cases, gradient norms exponentially decay with system size, $\|\partial C / \partial \theta\| \sim e^{-\alpha n}$, leading to the inability of gradient-based methods to work effectively in case of increasing qubits; a property that has been proven rigorously in McClean et al. [20] and has been mentioned in Cerezo et al. [15]. As standard quantum gates (Pauli, CNOT, Rotation) can be associated with the Clifford group and n -qubit rotors are generalized uniformly regardless of the system size (Section 3.2), using variational parameters restricted to the rotor (Lie-group) manifold and optimizing geodesically becomes a natural solution to the problem; since it ensures the gradients remain dependent on the same spectrum conditions used in Section 3.3 rather than an exponential flattening of the unconstrained landscape [15][20]. It is merely a theoretical proposal generalizing the concept proposed by McClean et al. [20] to Clifford structured ansatz; the comparison of polynomial versus exponential decay has not been re-benchmarked in this section.

7. Discussion and Conclusion

The pattern in Sections 4–6 is that all of the mathematical truths mentioned have in common that $\mathbb{C} \cong \mathbb{C}l^{even}(2)$, $\mathbb{H} \cong \mathbb{C}l^{even}(3)$, and all other hypercomplex systems in applied computations can be embedded in a rotor subalgebra of some “ $Cl(n)$ ”[1][2][3]. The reason why this isn't just classification nicety is that an algorithm stated in Clifford terms (CGENN layer, QLMS filter, rotor constraint in

variationalansatz) can generalize to n dimensions simply by adding new grades without changing its structure since there is a common axiom ($v^2 = Q(v, v)$) for all dimensions [2][14].

Open questions: The price of general multi-vector arithmetic is $O(2^n)$, while the theory of sparse representations limited only to lower grades (0,1,2) used practically in all tasks is still underdeveloped [3][14]. Generalization of CGENN-type architectures for sequential/recurrent geometric data (trajectories, videos) through bivector-valued hidden layers is not well-developed yet [2][3]. The principled, as opposed to heuristical, choice of signature (p, q) of algebra for each particular problem and development of closer quantum-classical hybrids using Clifford structure as their shared language are open [3][2][16][19].

3. **Conclusion:** Geometric constraints are algebraic in Clifford algebra, thus algorithms using such structures inherit them automatically from the very structure instead of learning them. It is possible that the principle above can be taken to explain the parameter efficiency in equivariant networks [3] and convergence behavior in hypercomplex adaptive filtering [8]; it also motivated (but didn't prove as of Section 6) improved trainability in variational quantum circuits [20][15]. With geometrical data becoming more relevant to machine learning and computing in general, Clifford algebra provides a solid functional-analytical basis (Sections 2–3) and dimension-independent architecture language (Sections 4–6).

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